TOP-Q — Day 1 worked notes

Coupled-mode theory → SSH Hamiltonian → Bloch bands → validated code

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22 October 2026

How to use these notes. This is a fully worked reference. It is only useful if you can reproduce it: close the PDF and re-derive Sections 1 and 2 on paper, then explain each test in Section 3 aloud. Evidence labels used throughout: [Derived] analytic result; [Assumption]; [Verified-by-test]; [Simulated] numerical observation; [Literature]; [Hypothesis].

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1 Block 1 — Deriving the coupled-mode equations

1.1 Physical picture

A single-mode dielectric waveguide confines light in its core, but the mode field does not stop at the core boundary: it decays *evanescently* into the cladding. Place a second identical waveguide a gap g away and the tail of mode 1 overlaps core 2. That overlap acts as a weak, distributed source for mode 2, so power transfers

back and forth along z. Coupled-mode theory (CMT) keeps the *shape* of each isolated mode and lets only its complex amplitude $a_n(z)$ evolve.

1.2 Assumptions [Assumption]

A1 Monochromatic field, angular frequency ω, time dependence $e^{-i\omega t}$; forward propagation $e^{+i\beta z}$.

A2 Scalar, paraxial, weakly guiding: one polarisation, so $E(x, y, z)$ obeys the scalar Helmholtz equation.

A3 Identical single-mode waveguides: all isolated modes have the same β_0 and shape (translated).

A4 Weak coupling: $\kappa \ll \beta_0$; modal overlaps $\int e_n e_m \, dA$ $(n \neq m)$ negligible.

A5 Lossless dielectrics (real index): this makes H Hermitian. Loss is added deliberately on Day 5.

A6 z-invariant structure (straight parallel guides): κ independent of z.

1.3 Step 1: wave equation

For a weakly guiding medium the transverse field obeys

$$\nabla^2 E + k_0^2 n^2(x, y)\, E = 0, \qquad k_0 = \frac{2\pi}{\lambda} = \frac{\omega}{c} \;\; [\mathrm{m}^{-1}]. \tag{1}$$

$n^2(x, y)$ is the relative permittivity profile of the whole array.

1.4 Step 2: isolated waveguide m

Let $n_m^2(x, y)$ be the profile of cladding plus *only* core m. Its guided mode $e_m(x, y)e^{i\beta_0 z}$ satisfies

$$\nabla_t^2 e_m + \big(k_0^2 n_m^2 - \beta_0^2\big) e_m = 0, \qquad \nabla_t^2 = \partial_x^2 + \partial_y^2. \tag{2}$$

e_m is real and positive everywhere for a fundamental mode (no nodes) — this matters for the sign of κ.

1.5 Step 3: the array seen from waveguide m

Write the full profile as $n^2 = n_m^2 + \sum_{p \neq m} \Delta n_p^2$, where $\Delta n_p^2(x, y) = n_{\mathrm{core}}^2 - n_{\mathrm{cl}}^2$ inside core p and 0 elsewhere.

1.6 Step 4(a): substitute the ansatz

Ansatz: $E = \sum_m a_m(z)\, e_m(x, y)\, e^{i\beta_0 z}$. Using $\partial_z^2(a e^{i\beta_0 z}) = (a'' + 2i\beta_0 a' - \beta_0^2 a) e^{i\beta_0 z}$:

$$\begin{aligned} \nabla^2 E &= \textstyle\sum_m e^{i\beta_0 z} \big[a_m \nabla_t^2 e_m + e_m (a_m'' + 2i\beta_0 a_m' - \beta_0^2 a_m) \big], \\ k_0^2 n^2 E &= \textstyle\sum_m e^{i\beta_0 z}\, k_0^2 \big(n_m^2 + \sum_{p \neq m} \Delta n_p^2 \big) a_m e_m. \end{aligned}$$

Add them and replace $\nabla_t^2 e_m$ using (2): the terms $-(k_0^2 n_m^2 - \beta_0^2) a_m e_m$, $-\beta_0^2 a_m e_m$ and $+k_0^2 n_m^2 a_m e_m$ cancel exactly. What survives is [Derived]

$$\sum_m \Big[(a_m'' + 2i\beta_0 a_m')\, e_m + k_0^2\, a_m \sum_{p \neq m} \Delta n_p^2\, e_m \Big] = 0. \tag{3}$$

Interpretation: the isolated-mode physics cancels; only the "extra" index of the *other* cores drives changes in a_m.

1.7 Step 4(b): slowly varying envelope approximation (SVEA)

Coupling makes a_m vary on the length scale $1/\kappa$, so $a_m' \sim \kappa a_m$ and $a_m'' \sim \kappa^2 a_m$. Then

$$\frac{|a_m''|}{2\beta_0 |a_m'|} \sim \frac{\kappa}{2\beta_0} \ll 1 \quad \text{by A4},$$

so a_m'' is dropped. Physically: the envelope changes negligibly over one optical wavelength in the medium. [Assumption]

1.8 Step 4(c): project onto mode n

Multiply (3) (without a'') by e_n and integrate over the cross-section. By A4, $\int e_m e_n\, dA \approx \delta_{mn} \int e_n^2\, dA$:

$$2i\beta_0\, a_n' \int e_n^2\, dA = -k_0^2 \sum_m a_m \int \Big(\sum_{p\neq m} \Delta n_p^2 \Big) e_m e_n\, dA.$$

Divide by $2i\beta_0 \int e_n^2 dA$ and use $-1/i = +i$: [Derived]

$$\boxed{\frac{da_n}{dz} = i \sum_m \kappa_{nm} a_m, \qquad \kappa_{nm} = \frac{k_0^2}{2\beta_0} \frac{\int \big(\sum_{p\neq m} \Delta n_p^2\big) e_m e_n\, dA}{\int e_n^2\, dA}} \tag{4}$$

Nearest neighbours ($m = n \pm 1$): the integrand needs e_n to be large, i.e. we are inside core n, so the dominant term of $\sum_{p\neq m}$ is $p = n$:

$$\kappa_{n,n\pm1} \simeq \frac{k_0^2}{2\beta_0} \frac{\int_{\text{core } n} \Delta n_n^2\, e_n\, e_{n\pm1}\, dA}{\int e_n^2\, dA}. \tag{5}$$

Non-nearest neighbours involve $e_{n\pm2}$ inside core n, which is exponentially smaller (Section 1.10) — this is the nearest-neighbour approximation [Assumption].

1.9 Matrix form and power conservation

Collect amplitudes in $\boldsymbol{a} = (a_1, \ldots, a_M)^T$: $d\boldsymbol{a}/dz = iH\boldsymbol{a}$, $H_{nm} = \kappa_{nm}$ (off-diagonal) and $H_{nn} = \delta\beta_n$ (on-site detuning + self-shift). Then

$$\frac{d}{dz}\, \boldsymbol{a}^\dagger \boldsymbol{a} = \boldsymbol{a}^\dagger \big(-iH^\dagger + iH \big) \boldsymbol{a} = 0 \quad \text{iff } H = H^\dagger.$$

So lossless $\Leftrightarrow$ Hermitian $H \Leftrightarrow \sum_n |a_n|^2$ conserved [Derived]. (For $|a_n|^2$ to *be* power in watts, e_n must be normalised so that $\frac{\beta_0}{2\omega\mu_0} \int e_n^2 dA = 1$; the dynamics do not depend on this choice.) Solution: $\boldsymbol{a}(z) = e^{iHz}\boldsymbol{a}(0)$. If $H\boldsymbol{\psi} = b\boldsymbol{\psi}$, then a launched supermode evolves as $\boldsymbol{\psi} e^{ibz}$, i.e. total phase $e^{i(\beta_0+b)z}$: **eigenvalue b = propagation-constant shift.**

1.10 Answers to the Block-1 questions

Q1 (units). $[k_0^2] = \mathrm{m}^{-2}$, $[\beta_0] = \mathrm{m}^{-1}$, Δn^2 dimensionless, and the ratio of integrals is dimensionless (same $e^2\, dA$ in numerator and denominator). So $[\kappa] = \mathrm{m}^{-2}/\mathrm{m}^{-1} = \mathrm{m}^{-1}$ (rad/m). ✓

Q2 (sign). Inside core n: $\Delta n_n^2 > 0$ (core index exceeds cladding), $e_n > 0$, $e_{n\pm1} > 0$ (fundamental modes have no nodes). The integrand is positive, so $\kappa > 0$. With the opposite convention $e^{-i\beta z}$, step 4(c) gives $da/dz = -iHa$ with the same positive κ; equivalently $H \to -H$, so $b \to -b$. Because the SSH spectrum is symmetric about zero (Section 2.6) the band *shapes* are unchanged, but the meaning of "$b > 0$" (faster vs. slower phase) flips. Always state your convention — that is why it is fixed in the conventions sheet.

Q3 (self-shift). The $m = n$ term of (4) gives $\sigma_n = \frac{k_0^2}{2\beta_0} \int \sum_{p\neq n} \Delta n_p^2 e_n^2\, dA / \int e_n^2 dA$: the neighbours' cores sit in mode n's own tail. Every *interior* guide has two neighbours, so the same σ; we absorb it into β_0. The **two end guides have only one neighbour**, so their self-shift is $\approx \sigma/2$ smaller: an *on-site detuning localised exactly where the edge mode lives.* Because $\sigma \propto (e^{-\gamma g})^2$ while $\kappa \propto e^{-\gamma g}$, it is second order, $\sigma/\kappa \sim e^{-\gamma g} \ll 1$, but it is a genuine chiral-symmetry-breaking term in a real device. **Record as Limitation L1.**

Q4 (exponential gap dependence). Outside its core, mode m satisfies $\nabla_t^2 e_m = (\beta_0^2 - k_0^2 n_{\text{cl}}^2) e_m \equiv \gamma^2 e_m$, so it decays as $e^{-\gamma r}$ (slab: exactly $e^{-\gamma|x|}$; 2-D: $K_0(\gamma r) \sim e^{-\gamma r}/\sqrt{r}$). Inside core n, a distance $\approx g$ from the edge of core m, $e_m \propto e^{-\gamma g}$, hence

$$\kappa(g) \approx \kappa_0 e^{-\gamma g} = \kappa_0 e^{-g/g_0}, \qquad g_0 = \gamma^{-1}, \qquad \frac{\delta\kappa}{\kappa} = -\gamma\, \delta g.$$

A fixed gap error δg therefore produces a fixed *fractional* coupling error. That is the physical justification for **multiplicative** coupling disorder $\kappa \to \kappa(1+\epsilon)$ (Day 3) [Derived].

2 Block 3 — SSH equations, Bloch Hamiltonian, bands

2.1 Real-space SSH equations

Alternate the gaps $g_1, g_2, g_1, \ldots$ so the couplings alternate κ_1, κ_2. Group guides in unit cells $n = 1, \ldots, N$, each with an A and a B guide; κ_1 couples $A_n \leftrightarrow B_n$ (intra-cell), κ_2 couples $B_n \leftrightarrow A_{n+1}$ (inter-cell). With $\delta\beta_n = 0$ (identical guides), (4) gives [Derived]

$$\frac{da_n}{dz} = i(\kappa_1 b_n + \kappa_2 b_{n-1}), \qquad \frac{db_n}{dz} = i(\kappa_1 a_n + \kappa_2 a_{n+1}). \tag{6}$$

Key structure: A couples only to B, B only to A (*bipartite* lattice). Real-space H for $M = 6$:

$$H = \begin{pmatrix} 0 & \kappa_1 & & & & \\ \kappa_1 & 0 & \kappa_2 & & & \\ & \kappa_2 & 0 & \kappa_1 & & \\ & & \kappa_1 & 0 & \kappa_2 & \\ & & & \kappa_2 & 0 & \kappa_1 \\ & & & & \kappa_1 & 0 \end{pmatrix}.$$

2.2 Bloch's theorem on a lattice

In an infinite (or periodic) array, H commutes with the one-cell translation $T : (a_n, b_n) \mapsto (a_{n+1}, b_{n+1})$. Commuting operators share eigenvectors, and T is unitary, so its eigenvalues are e^{ik}. Hence eigenmodes take the form $a_n = a\, e^{ikn}$, $b_n = b\, e^{ikn}$, with $k = q\Lambda$ dimensionless (Λ = transverse cell period = centre-to-centre distance over two guides) and $k \in (-\pi, \pi]$ (the Brillouin zone; k and $k + 2\pi$ are the same state).

2.3 Bloch Hamiltonian

Substitute into (6): $b_{n-1} = b\, e^{ik(n-1)}$, $a_{n+1} = a\, e^{ik(n+1)}$; cancel e^{ikn}:

$$\frac{d}{dz}\begin{pmatrix} a \\ b \end{pmatrix} = i\, H(k) \begin{pmatrix} a \\ b \end{pmatrix}, \qquad \boxed{H(k) = \begin{pmatrix} 0 & h(k) \\ h^*(k) & 0 \end{pmatrix}, \quad h(k) = \kappa_1 + \kappa_2 e^{-ik}} \text{ [Derived]}$$

2.4 Bands

$\det(H(k) - b\mathbb{1}) = b^2 - |h|^2 = 0$, and $|h|^2 = (\kappa_1 + \kappa_2 \cos k)^2 + \kappa_2^2 \sin^2 k = \kappa_1^2 + \kappa_2^2 + 2\kappa_1\kappa_2 \cos k$:

$$\boxed{b_\pm(k) = \pm\sqrt{\kappa_1^2 + \kappa_2^2 + 2\kappa_1\kappa_2 \cos k}} \text{ [Derived]} \tag{7}$$

- Maximum of $|b|$ at $k = 0$: $\kappa_1 + \kappa_2$ (total band extent $2(\kappa_1 + \kappa_2)$).
- Minimum at $k = \pi$: $|\kappa_1 - \kappa_2|$. **Gap** $= 2|\kappa_1 - \kappa_2|$, centred on $b = 0$, i.e. on β_0.
- Gap closes only for $\kappa_1 = \kappa_2$ (uniform array; folded band $b = \pm 2\kappa \cos(k/2)$).
- Physical meaning of the slope: $\partial b/\partial k$ is the transverse group velocity (how fast light walks sideways per unit z, in cells); its spread is *discrete diffraction*.

2.5 *d*-vector form and what a σ_z term would mean

Any 2×2 Hermitian matrix is $d_0\mathbb{1} + \boldsymbol{d} \cdot \boldsymbol{\sigma}$. Matching $h = d_x - i d_y$:

$$d_x(k) = \kappa_1 + \kappa_2 \cos k, \qquad d_y(k) = \kappa_2 \sin k, \qquad d_z = d_0 = 0, \qquad b_\pm = \pm|\boldsymbol{d}|.$$

- $d_z\sigma_z$ = **opposite** detunings on A and B: $\delta\beta_A = +d_z$, $\delta\beta_B = -d_z$ (alternating guide widths) — the Rice–Mele model. It opens a gap even at $\kappa_1 = \kappa_2$ and moves states off $b = 0$.
- $d_0\mathbb{1}$ = a **common** shift of every guide's β (e.g. a uniform width error): shifts all b by d_0.
- With $d_z = d_0 = 0$: $\sigma_z H(k) \sigma_z = -H(k)$. This is **chiral (sublattice) symmetry**; it forces eigenvalues in $\pm b$ pairs. Both perturbations above break it. (Developed on Day 2.)

2.6 Why $\pm\delta$ give identical bulk bands

Parametrise $\kappa_1 = \bar{\kappa}(1-\delta)$, $\kappa_2 = \bar{\kappa}(1+\delta)$. Equation (7) is symmetric under $\kappa_1 \leftrightarrow \kappa_2$ ($\delta \to -\delta$). So **the bulk spectrum cannot distinguish the two dimerisations**. They differ only in how the chain is *cut*: an A-terminated chain ends on a weak bond if $\kappa_1 < \kappa_2$ ("dangling" end guide) and on a strong bond if $\kappa_1 > \kappa_2$. Fig. 1(b) shows the consequence.

Correction to my earlier instructions. On 21 Oct I wrote that the bands for $\kappa_1/\kappa_2 = 0.5$ and 2, plotted in units of κ_2, are identical. That is wrong as stated: in units of κ_2 the second case is the first scaled by 2. They are identical when the *mean* coupling $\bar{\kappa}$ is held fixed (equivalently, under $\kappa_1 \leftrightarrow \kappa_2$). The figure below uses this fixed-$\bar{\kappa}$ parametrisation.

2.7 Finite rings: which k are allowed

A ring of N cells requires $a_{n+N} = a_n \Rightarrow e^{ikN} = 1 \Rightarrow k_m = 2\pi m/N$, $m = 0, \ldots, N-1$. Its $2N$ eigenvalues are exactly $b_\pm(k_m)$ — no approximation. This is what makes the ring a sharp numerical test of (7). If N is even, $k = \pi$ is allowed, so the ring gap equals $2|\kappa_1 - \kappa_2|$ exactly.

3 Block 4 — Implementation and validation

3.1 Module design (`topq/ssh.py`)

Function	Purpose / design decision
`ssh_couplings(M,k1,k2)`	$[\kappa_1, \kappa_2, \kappa_1, \ldots]$, length $M-1$; chain always starts on A with κ_1 (conventions sheet).
`hamiltonian_from_couplings`	General tridiagonal builder from an arbitrary coupling array + optional on-site array + optional ring closure. Day 3 disorder will pass perturbed arrays here, so disorder and clean code share one builder.
`ssh_hamiltonian(M,k1,k2,...)`	Thin wrapper; ring requires even M (otherwise the closure would join A to A and break the cell structure).
`bloch_hamiltonian(k,...)`	Vectorised 2×2 $H(k)$, shape `k.shape+(2,2)`.
`bloch_bands(k,...)`	Eq. (7); `clip` protects $\sqrt{\cdot}$ from -10^{-17} round-off at $\kappa_1 = \kappa_2, k = \pi$.

Why `eigh`, not `eig`: for Hermitian matrices `numpy.linalg.eigh` uses a symmetric solver: real sorted eigenvalues, orthonormal eigenvectors, backward-stable. `eig` may return tiny spurious imaginary parts and non-orthogonal vectors for degenerate eigenvalues. (Once loss is added on Day 5 H is non-Hermitian and we must switch to `eig` deliberately.)

Units: the code is unit-agnostic. Whatever unit you pass κ_1, κ_2 in (e.g. mm^{-1}), b comes out in the same unit.

3.2 Test suite (`tests/test_ssh.py`) — 23 tests, all pass **[Verified-by-test]**

Test	What it proves (and why it is independent)
Ring vs. analytic	Real-space ring eigenvalues equal (7) at k_m to $< 10^{-12}$ for 4 coupling pairs. Independent: real-space matrix vs. Bloch algebra.
Real symmetric	Lossless H is Hermitian (Section 1.9).
Spectrum symmetric	Open chains ($M = 40, 41$) have $\{b\} = \{-b\}$: numerical signature of chiral symmetry.
Ring gap	Gap $= 2|\kappa_1 - \kappa_2|$ (even N contains $k = \pi$).
Bad inputs	Negative coupling, odd ring, wrong on-site length, NaN all raise `ValueError`.
Bloch consistency	$H(k)$ Hermitian, 2π-periodic, `eigvalsh`$(H(k)) = b_\pm(k)$.
Uniform open chain	$\kappa_1 = \kappa_2$: $b_j = 2\kappa\cos\frac{\pi j}{M+1}$ — a textbook result derived without Bloch theory.
Fully dimerised	$\kappa_2 = 0$: isolated dimers $\pm\kappa_1$. $\kappa_1 = 0$: dimers $\pm\kappa_2$ *plus two isolated end guides at exactly* $b = 0$ — the extreme limit of the edge states.

Do the tests have teeth? (mutation test). I deliberately injected three bugs and re-ran the suite: closing the ring with κ_1 instead of κ_2 (6 failures), $\cos k \to \sin k$ in the bands (5 failures), all couplings set to κ_1 (5 failures). A test suite that cannot fail is not evidence; this one catches realistic mistakes.

3.3 Independent momentum assignment for the figure

Plotting ring eigenvalues at the analytic k_m would be circular. Instead, for each real eigenvector $\boldsymbol{\psi}$ of the ring, compute $\langle\boldsymbol{\psi}|T|\boldsymbol{\psi}\rangle$ with T = shift by two sites. A real eigenvector is an equal mixture of $\pm k$ Bloch states, so $\langle T\rangle = \frac{1}{2}(e^{ik} + e^{-ik}) = \cos k$, giving $|k| = \arccos\langle T\rangle$ *from the eigenvector alone*. Point-wise agreement with (7): $\leq 2 \times 10^{-15}$ for $\delta = \pm 1/3$. For $\delta = 0$ it is 3×10^{-8}: not a bug, but the ill-conditioning of arccos near ± 1 ($\delta k \sim \sqrt{\epsilon}$ for round-off $\epsilon \sim 10^{-16}$) plus the band touching at $k = \pi$. [Simulated]

4 Block 5 — Figure, observations, reading

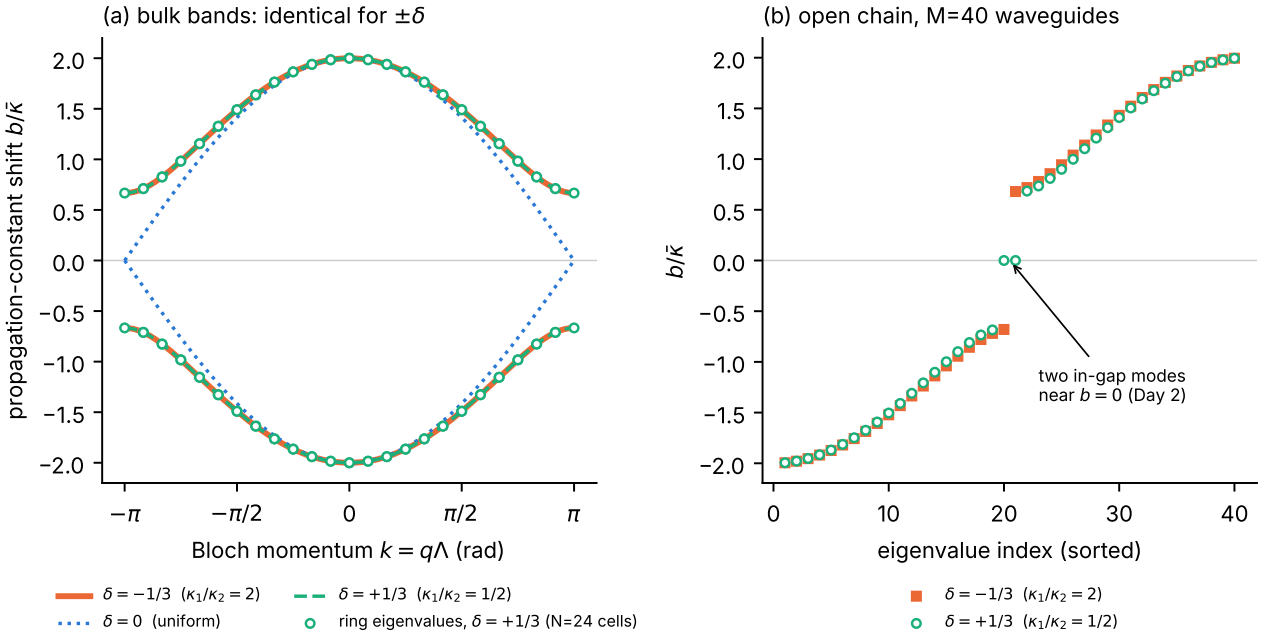


Figure 1: (a) Bulk bands (7) for $\delta = -1/3, 0, +1/3$ at fixed $\bar{\kappa}$; circles: real-space ring eigenvalues ($N = 24$ cells, $\delta = +1/3$) with momenta assigned from eigenvectors. (b) Spectrum of an open chain of $M = 40$ guides for $\delta = \pm 1/3$. Generated by `scripts/day1_bands.py` from `data/day1_bands.npz`; deterministic, no random numbers.

Observations [Simulated]

1. The $\delta = \pm 1/3$ bulk bands coincide; ring eigenvalues lie on them to 10^{-15}.
2. Gap: $2|\kappa_1 - \kappa_2|/\bar{\kappa} = 4/3$; band edges at $\pm 2/3$ and ± 2 (in units of $\bar{\kappa}$), as predicted.
3. Open chain: $\delta = +1/3$ ($\kappa_1 < \kappa_2$) has **two** eigenvalues inside the gap at $|b|/\bar{\kappa} = 9.54 \times 10^{-7}$; $\delta = -1/3$ has none (smallest $|b|/\bar{\kappa} = 0.680$, just above the bulk edge $2/3$ because of finite size).
4. The two in-gap values are not exactly 0: they are $\pm$-split. Their magnitude equals $(\kappa_1/\kappa_2)^N = (1/2)^{20} = 9.537 \times 10^{-7}$ to the digits shown — a strong hint of exponential hybridisation of the two edge modes. [Hypothesis — to be derived on Day 2, not yet claimed.]

4.1 Reading notes: Su, Schrieffer & Heeger

Citation. W. P. Su, J. R. Schrieffer, A. J. Heeger, "Solitons in Polyacetylene," *Phys. Rev. Lett.* **42**, 1698 (1979), doi:10.1103/PhysRevLett.42.1698. Long version: *Phys. Rev. B* **22**, 2099 (1980). [Literature — verify title/DOI on the APS page]

What it actually studies. Electrons in trans-polyacetylene, a carbon chain whose bonds spontaneously alternate (dimerise). The hopping depends on the lattice displacement, $t_{n+1,n} = t_0 - \alpha(u_{n+1} - u_n)$, so alternating displacements give alternating hoppings — the same tridiagonal alternating-coupling matrix as our array. **Two degenerate ground states** (A and B phases: the two bond-alternation patterns). A domain wall between them is a **soliton** carrying an electronic state at mid-gap ($E = 0$). The 1980 paper reports a soliton half-width $l \approx 7$ lattice constants and formation energy ≈ 0.42 eV for a 1.4 eV gap, and the reversed spin–charge relations (neutral: spin $\frac{1}{2}$; charged: spin 0). **What it does *not* contain.** No winding number,

no "topology" language, no photonics. Those come later (e.g. Ozawa *et al.* 2019 review). **Do not cite SSH 1979 for topological invariants. Mapping to TOP-Q.** Electron hopping $t \leftrightarrow$ coupling κ; energy $E \leftrightarrow$ propagation-constant shift b; time $\leftrightarrow z$; mid-gap soliton state $\leftrightarrow$ edge/domain-wall mode at $b = 0$. Differences: our "hoppings" are fixed by fabrication (no lattice dynamics, no electron–phonon feedback), photons are non-interacting bosons, and our system is driven by a launched field rather than filled to a Fermi level.

5 Answers to the four Day-1 check questions

1. Why multiplicative coupling disorder? Because $\kappa \propto e^{-\gamma g}$ (Q4 above): $\delta\kappa/\kappa = -\gamma\delta g$. It rests on A3 (identical modes, so a single γ) and the evanescent-tail argument.

2. A uniform shift $\delta\beta$ of every guide. $H \to H + \delta\beta\mathbb{1}$ shifts every b by $\delta\beta$, including the edge mode: it is *not* pinned against this, because $\mathbb{1}$ breaks chiral symmetry ($\Gamma\mathbb{1}\Gamma = +\mathbb{1}$). But it adds $\delta\beta L$ equally to two arms that share it, so the *relative* phase of an MZI is unchanged. What matters for an interferometer is the *difference* in shifts between arms. This is precisely risk R2 and why Day 6 measures differential phase. [Derived]

3. What swapping $\kappa_1 \leftrightarrow \kappa_2$ changes physically. Not the interior — the bulk bands are identical — but the termination: whether the last guide hangs on a weak or a strong bond. Fig. 1(b): weak-bond termination $\Rightarrow$ in-gap states at $b \approx 0$; strong-bond termination $\Rightarrow$ none. This is the seed of bulk–boundary correspondence (Day 2).

4. The $1/\kappa$ length scale. $\pi/(2\kappa)$ is the full transfer length of two coupled guides; $1/\bar{\kappa}$ sets the discrete-diffraction length. To separate an edge mode from the bulk by propagation you need the phase difference $\Delta b\, L \gtrsim \pi$ between the edge mode and the nearest bulk state, i.e. $L \gtrsim \pi/|\kappa_1 - \kappa_2|$ (half the gap sets the beat). Real numbers for κ and L will be extracted from Blanco-Redondo *et al.* (2016) on Day 2 — **not assumed**.

6 Assumptions and limitations recorded today

ID	Type	Statement
A1–A6	Assumption	Listed in Section 1.2.
L1	Limitation	End-guide self-shift differs by $\sim \sigma/2$: an edge-localised on-site detuning that breaks chiral symmetry in real arrays (second order in $e^{-\gamma g}$).
L2	Limitation	Next-nearest-neighbour coupling neglected; it couples A–A and B–B and breaks chiral symmetry.
L3	Limitation	Scalar/single polarisation; no dispersion (single λ).
L4	Limitation	Non-orthogonality of modes neglected (A4); rigorous CMT treats it (candidate refs: A. Yariv, IEEE JQE 9, 919 (1973); W.-P. Huang, JOSA A 11, 963 (1994) — *unverified, check before citing*).

7 Self-test before Day 2 (answer without notes)

1. Which single assumption makes $\sum |a_n|^2$ conserved, and which line of the derivation uses it?
2. Why does the SVEA fail if κ approaches β_0? What physical structure would that be?
3. Write $H(k)$ for a chain whose unit cell starts on B instead of A. Do the bands change? Does $h(k)$?
4. Why must a ring have an even number of guides, and what would an odd ring physically be?
5. Explain in two sentences why the ring test is stronger evidence than plotting (7).